

Week 7 - 8 Assignment Part 2

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Abstract

This report outlines the process of integrating a TurtleBot3 robot with a Gazebo simulation environment and using only the camera feed for autonomous navigation through a track. The primary objective was to use computer vision techniques for detecting the track and computing the robot's velocity commands (linear and angular) to successfully traverse the path. The approach employed curvature estimation based on lane detection and path curvature.

1. Introduction

The task involved using the **TurtleBot3** robot in a **Gazebo** environment with the camera as the only available sensor. The main goal was to navigate a track within the world using visual input, making decisions based solely on camera data. This challenge required the integration of multiple ROS2 packages, manipulation of Gazebo worlds, and implementation of a computer vision-based method to control the robot's movements.

Objective

- **Test the Robot Model:** Ensure that the Gazebo world loads properly and integrates with TurtleBot3.
- **Camera-based Navigation:** Generate velocity commands using the camera feed for autonomous navigation.
- **Task Requirements:** Complete the task using only the camera as the sensor to navigate the track.

2. Methodology

2.1 Testing the Model

To begin, the **robot_world.zip** file was downloaded, extracted, and placed in the home directory. This folder structure contained the model configuration and world files needed for the task:

- **robot_world/**
 - **models/**
 - **track1/**
 - **model.config**
 - **model.sdf**
 - **Track_world.world**

2.3 Camera-based Navigation

The core task was to navigate the robot using only its camera. The chosen method for generating the velocity commands was curvature estimation, which involved detecting the track and calculating the curvature to control the robot's movement. The steps followed were:

1. **Camera Feed Subscription:**
The camera feed was accessed through the `/camera/image_raw` topic. Images from this topic were processed to extract lane markers.
2. **Lane Detection:**
Using OpenCV, lane markers were detected using edge detection and Hough Transform. The curvature of the detected lanes was computed based on the camera feed.
3. **Velocity Calculation:**
A fixed linear velocity was applied, and the corresponding angular velocity was calculated based on the curvature. The formula used was:
$$\text{angular_velocity} = \frac{\text{linear_velocity}}{\text{curvature}}$$

This allowed the robot to follow the path.
4. **Publishing Commands:**
The calculated linear and angular velocities were published to the `cmd_vel` topic to control the robot's movement.

3. Results

5. Conclusion

In this lab, the TurtleBot3 was successfully integrated into a custom Gazebo world, with navigation achieved using only the camera. The method of curvature estimation worked

effectively for navigating the track. This task demonstrated the potential of computer vision techniques in robotics and the importance of sensor-based decision-making for autonomous navigation.





